Reading the Decoder: A Distribution-Free Streaming Certificate for an LLM's Effective Decoding Temperature

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Abstract

An autoregressive language model is trained to *predict* text, so when it *generates*, it has an incentive to model the decoder that turns its logits into tokens (a temperature, a top-k cutoff, a banned-token list). This write-up reports preliminary steps toward reading that decoder off the model's own logits, training-free. Our central result is a **partial proof**: a streaming estimator built from a grid ("bucket") of fixed temperature experts is exactly Vovk's Aggregating Algorithm for log loss, and therefore carries two *per-sequence, distribution-free* guarantees on arbitrary token streams — a forecast-regret bound of $\ln G$ against the best bucket in hindsight, and a convex level-set error bar around the grid MLE that needs no distributional assumptions (Lemma 1). We then report three preliminary findings, framed as questions: **(1)** the model tracks a token blacklist and a top-k cutoff from its own constrained text; **(2)** a streaming read-out of $\hat{\beta}_t$ predicts known-temperature self-text near-optimally but finds little to recalibrate on natural text, where Qwen-2.5-3B is temperature-robust; and **(3)** closed-loop correction self-neutralizes, so using the read-out to *generate* better is still open. All numbers are script-generated and cited by name.

1 The research question

At each step an autoregressive language model emits a distribution over next tokens; *decoding* chooses how to sample from it. Because models are trained to predict text rather than to generate it, a model that reads back its own generated prefix has an incentive to infer the decoder that produced it and adjust its predictions accordingly (Basu et al., 2020). The question this project addresses is whether that decoder — concretely, an effective inverse-temperature $\beta = 1/T$ — can be *read off the model's own logits online, training-free, and with guarantees*, and whether the read-out is useful for prediction or generation. This sits at the intersection of post-hoc calibration (Guo et al., 2017; Liu et al., 2023) and the algorithms-with-predictions viewpoint (Mitzenmacher and Vassilvitskii, 2020) of the course, in which a possibly-unreliable prediction augments a downstream decision; what we then most want is a *worst-case, per-sequence* guarantee that holds even when the stream is misspecified. Section 2 delivers one; Section 3 reports three preliminary experiments, with supporting machinery (a fast point filter, a corrector, control experiments) in the appendices.

2 The bucket streaming certificate

Fix a grid (a set of "buckets") of inverse-temperatures $\beta_1 < \cdots < \beta_G$, log-spaced over a bounded range. Treat each β_g as a fixed *expert* that forecasts the next token with probability $P_g(x_t) = \mathrm{softmax}(\beta_g\, \ell_t)[x_t]$ on the model's *raw* logits $\ell_t \in \mathbb{R}^V$ (no rescaling — fixed experts are what make the guarantees hold). Let

$$L_n(\beta) = \sum_{t \le n} \big[\mathrm{lse}(\beta \ell_t) - \beta\, \ell_t[x_t]\big], \qquad \mathrm{lse}(\cdot) = \log \textstyle\sum_v \exp(\cdot),$$

be the cumulative log loss of bucket β, let π_0 be a prior on the grid, and let the streaming forecaster predict x_t with probability $\sum_g w_g(t)\, P_g(x_t)$, where $w_g(t) \propto \pi_0[g]\, e^{-L_{t-1}(\beta_g)}$ is the Bayes posterior over buckets. This is precisely Vovk's Aggregating Algorithm for log loss (Vovk, 1998; Cesa-Bianchi and Lugosi, 2006), and the following holds for *every* logit/token sequence — with no assumption that tokens were sampled from any P_β, or from anything at all.

Lemma 1 (per-sequence certificate)**.** *For any logit sequence $\ell_{1:n}$ and any token sequence $x_{1:n}$: (i) the forecaster's cumulative log loss is $-\ln \sum_g \pi_0[g]\, e^{-L_n(\beta_g)} \le L_n(\beta_g) + \ln(1/\pi_0[g])$ for every g; with uniform π_0 the regret against the best bucket in hindsight is at most $\ln G$. (ii) With uniform π_0 the posterior mode equals* $\arg\min_g L_n(\beta_g)$*, the best-fit (*effective*) inverse temperature on the grid, exactly. (iii) L_n is convex with $L_n''(\beta) = \sum_t \mathrm{Var}_{q_{\beta,t}}[\ell_t]$, $q_{\beta,t} = \mathrm{softmax}(\beta \ell_t)$, so for any $c > 0$ the sub-level set $\{\beta : L_n(\beta) \le \min L_n + c\}$ is an interval containing the sequence MLE; its endpoints, read off the stored grid loss profile, are a data-dependent error bar that needs no distributional assumptions.*

Proof. (i) The forecaster's per-step log probabilities telescope:

$$\textstyle\sum_t \ln \sum_g w_g(t) P_g(x_t) = \ln \sum_g \pi_0[g]\, e^{-L_n(\beta_g)} \ge \ln\big(\pi_0[g]\, e^{-L_n(\beta_g)}\big) \text{ for each } g.$$

(ii) Immediate from $w_g(n) \propto e^{-L_n(\beta_g)}$. (iii) Each summand of L_n is an lse of an affine function of β minus an affine function, hence convex; differentiate twice. □

"Distribution-free" here does double duty. Over the data process, the lemma assumes nothing about how $x_{1:n}$ was produced. Over the estimator's own assumptions, with uniform π_0 the mode in (ii) *is* the grid MLE and the bar in (iii) is an exact profile-likelihood interval (not a Gaussian approximation); the prior survives only as the constant $c = \ln G$. We report the mode, not the posterior mean: by (ii) it equals the best-fit bucket with zero slack, whereas the mean of the (unimodal but skewed — e^{-L_n} is log-concave by (iii)) posterior over a log-spaced grid is mildly biased.

Three honest caveats. First, the certificate is relative to the in-hindsight best-fit β, not the dial a counterparty actually set; under a misspecified decoder (a top-k cutoff, a blacklist) no single "true" β describes the stream, and the best-fit value is exactly what "effective temperature" should mean there. The bar then certifies *fit to the observed stream*, and a narrow bar means β is *identifiable*, not that any temperature fits well. Second, the interval in (iii) widens exactly when peaked logits make β unidentifiable ($\mathrm{Var}_q[\ell_t] \to 0$), so forecasting regret and estimation precision degrade in opposite directions. Third, the guarantee attaches to the mixture, not a point estimator (proper online predictors face an exp-concavity obstruction that improper mixtures avoid (Foster et al., 2018)), so transferring it to the fast filter (Appendix A) is an empirical, pre-registered claim.

Pre-registered validation. We committed four hypotheses, with sample sizes and thresholds, *before* generating any data (`paper/grid_certificate_prereg.md`); the bar uses $c = \ln G$ on a 121-bucket grid spanning $\log \beta \in [-3, 3]$. Table 1 reports the outcome. The bar covered the continuum MLE on every well-specified synthetic stream (200/200); on real Qwen-2.5-3B streams at β=1 the

pre-registered hypothesis	outcome	test	verdict
H1 coverage (synthetic, well-spec.)	200/200 cover the MLE	$p = 1.00$	passes
H2 non-vacuousness (Qwen, β=1)	median width 0.05 (< 0.2 in $\log\beta$)	$p < 10^{-9}$	holds
H3 degradation (peaked vs. natural)	median 0.45 vs. 0.25	$p < 10^{-67}$	holds
H4 transfer (faithful Laplace in bar)	88/96 inside	$p = 0.76$	passes

Table 1: Pre-registered evaluation of the certificate (hypotheses, sample sizes, and thresholds committed before any data were generated). Real streams are Qwen-2.5-3B, 32 prompts, 1000 tokens.

median width was 0.05 in $\log\beta$ (the grid-resolution floor — the bar is far from vacuous where logits carry temperature information) and widened on peaked-logit streams exactly as (iii) predicts. The transfer check to the fast filter passed (88/96 of estimates inside the bar), while the original β=1-linearized filter landed inside on only 10/96 — so the certificate extends to the corrected filter's outputs, not the original's.

3 Three preliminary findings

We frame the empirical work as three questions. All use Qwen-2.5-3B (base, fp16) and only the model's own logits; no training.

(1) Does the model model its decoder? Yes. We constrain the decoder while the model generates, then ask whether the model, reading back its own constrained text, has inferred the constraint. **Blacklist:** ban a token during generation; the model lowers its predicted probability on that token across the sequence ($\Delta p_{\text{late–early}} = -0.019$ for the top token) and below an unbanned control (-0.016, 95% CI $[-0.019, -0.013]$), with an early-position contrast ≈ 0 as a sanity check. **Top-k:** truncating to the top k produces no significant cliff at k itself; instead probability concentrates on the head, the more so the tighter the cutoff. So the model does track its decoder, though the top-k signal is broad sharpening rather than a k-specific edge — a conservative reading the controls do not let us strengthen to "infers the exact constraint."

(2) Can we use that to predict better? Yes on temperatured text, no in the wild. A streaming filter reads $\hat{\beta}_t$ off the prefix one token at a time (Appendix A). On Qwen self-text generated at a *known* temperature (64 sequences per T), $\hat{\beta}_t$ settles on the true $\log(1/T)$ from the prefix alone (bias -0.030 at T=0.7), and using it as a per-token temperature beats the default β=1 model by 0.108 nats at T=0.7 (up to 0.723 at T=2), within +0.025 nats of an oracle that *knows* $1/T$. But does *natural* text carry a recoverable temperature? Fitting the best global temperature per text type over 9 types from in-distribution to far-OOD (appendix Table 2), Qwen is temperature-robust everywhere: optimal $T \in [0.91, 1.08]$ and the best temperature buys $\leq +0.0032$ nats on any real type (+0.0283 only on shuffled gibberish). So the read-out is real and usable on temperatured text, but in the wild there is essentially nothing to recalibrate.

(3) Can we use it to generate better? Not yet. The tempting move is to feed $\hat{\beta}_t$ back into the sampler to "correct" the temperature as it generates. It self-neutralizes: correcting toward $\hat{\beta}$ drives the effective decoding temperature back to $\beta_{\text{dec}} \to 1$, quietly cancelling the very signal it reads off. We confirmed this across exact-grid filters, so it is not an approximation artifact. Turning the read-out into better generation is the honest open problem.

4 Limitations and next steps

These are preliminary steps, on a single model and tokenizer; the certificate certifies the effective β of the *observed* stream, never a counterparty's dial. The natural next step, for which it is the right tool, is misspecification: the *effective* temperature and goodness-of-fit that top-k and blacklist decoders register, where no single temperature reproduces the stream and the best-fit bucket is a projection (synthetic pilot: `scripts/e1_synthetic_pilot.py`).

A The fast point-estimate filter

The bucket certificate is a mixture forecaster; for a single-number read-out we use a companion streaming *Laplace* filter — a prior-regularized one-step online MLE on $\eta = \log \beta$. With a log-normal prior (mean 1, scale σ_0) and a per-token Newton/Fisher-scoring update of the categorical likelihood under $q = \mathrm{softmax}(\ell_t)$, the update uses $\mathrm{score}_t = \ell_t[x_t] - \mathbb{E}_q[\ell_t]$ and $J_t = \sigma_0^{-2} + \sum_{\tau \le t} \mathrm{Var}_q[\ell_\tau]$, with q evaluated at $\beta = 1$. This estimator is fast and is what produces the $\hat{\beta}_t$ used in finding (2); it carries *no* per-sequence guarantee (its experts are not fixed), which is exactly why the certificate of Section 2 exists and why we validate the filter *against* it (Table 1). Implementation and unit tests (shift invariance, batched = serial, recovery under matched-prior synthetic data) are in `src/decoding_decoding/counter_decode.py`.

B Corrector and supporting control experiments

A corrector $\beta_t^{\mathrm{dec}} = \beta^{\mathrm{target}} / \hat{\beta}_t^{p}$ divides a user's target by the estimate. An empirical sweep shows it pins the effective temperature in the "boredom" regime ($\beta^{\mathrm{target}} > 1$) but overshoots in the "confusion" regime ($\beta^{\mathrm{target}} < 1$), where low-$\beta$ self-generation degrades into incoherent text within $\sim$20 tokens (a garbage-in/garbage-out confound that limits what the confusion-regime numbers can claim). Two further controls sharpen the picture and are reported in full in the longer draft (`paper/decoding_paper.tex`): a sampler-switch experiment indicating the belief is not a fast per-token entropy read-out (though entropy and rank statistics are closely related, so this is a null on one specific mechanism, not a positive identification), and a closed-loop-vs-invariance comparison whose only firmly supported conclusion is "do not correct at $\beta^{\mathrm{target}} = 1$, because Qwen is already calibrated there."

text type	optimal T	ΔNLL vs β=1 (nats)	β=1 NLL	$H_{\beta=1}$
shuffled WikiText	1.08	+0.0283**	7.20	6.48
Recamán (A005132)	0.91	+0.0032**	0.08	0.14
ABC music	1.03	+0.0019**	2.48	2.34
WritingPrompts	1.04	+0.0011	2.87	2.74
Telugu	1.03	+0.0010**	1.12	1.06
Esperanto	1.03	+0.0007	2.72	2.63
Yoruba	1.00	+0.0001	3.06	3.11
WikiText	1.00	-0.0000	1.96	1.93
Gutenberg (archaic EN)	1.01	-0.0001	1.87	1.85

Table 2: Finding (2), natural text (full table): the best global temperature buys almost nothing over β=1 across 9 text types. ** marks a gain significant at the stated level; magnitudes are negligible on all real types.

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